A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2,3,4,!,2,5,I,2,3,4,5(No of Frames=3)'

Code:

```
#include <stdio.h>

int main() {
   int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
   int n = 12, frames = 3;
   int frame[3], time[3];
   int i, j, pos, hits = 0, faults = 0, t = 0;

   for(i = 0; i < frames; i++) {
      frame[i] = -1;
      time[i] = 0;
   }

   printf("Page Reference String: ");
   for(i = 0; i < n; i++)
      printf("%d ", pages[i]);

   printf("\nNumber of Frames: %d\n\n", frames);

   for(i = 0; i < n; i++) {
      int found = 0;

      for(j = 0; j < frames; j++) {
         if(frame[j] == pages[i]) {
            hits++;
            time[j] = ++t;
            found = 1;
            break;
         }
      }


}
```

```
if(!found) {
pos = 0;
for(j = 1; j < frames; j++) {
if(time[j] < time[pos])
pos = j;
}

frame[pos] = pages[i];
time[pos] = ++t;
faults++;
}
}

printf("\nPage Hits = %d\n", hits);
printf("Page Faults = %d\n", faults);
printf("Hit Ratio = %.2f%%\n", (hits/(float)n)*100);
printf("Fault Ratio = %.2f%%\n", (faults/(float)n)*100);

return 0;
```

Output:

```
Page Reference String: 1 2 3 4 1 2 5 1 2
    3 4 5
Number of Frames: 3



Page Hits = 2
Page Faults = 10
Hit Ratio = 16.67%
Fault Ratio = 83.33%
```